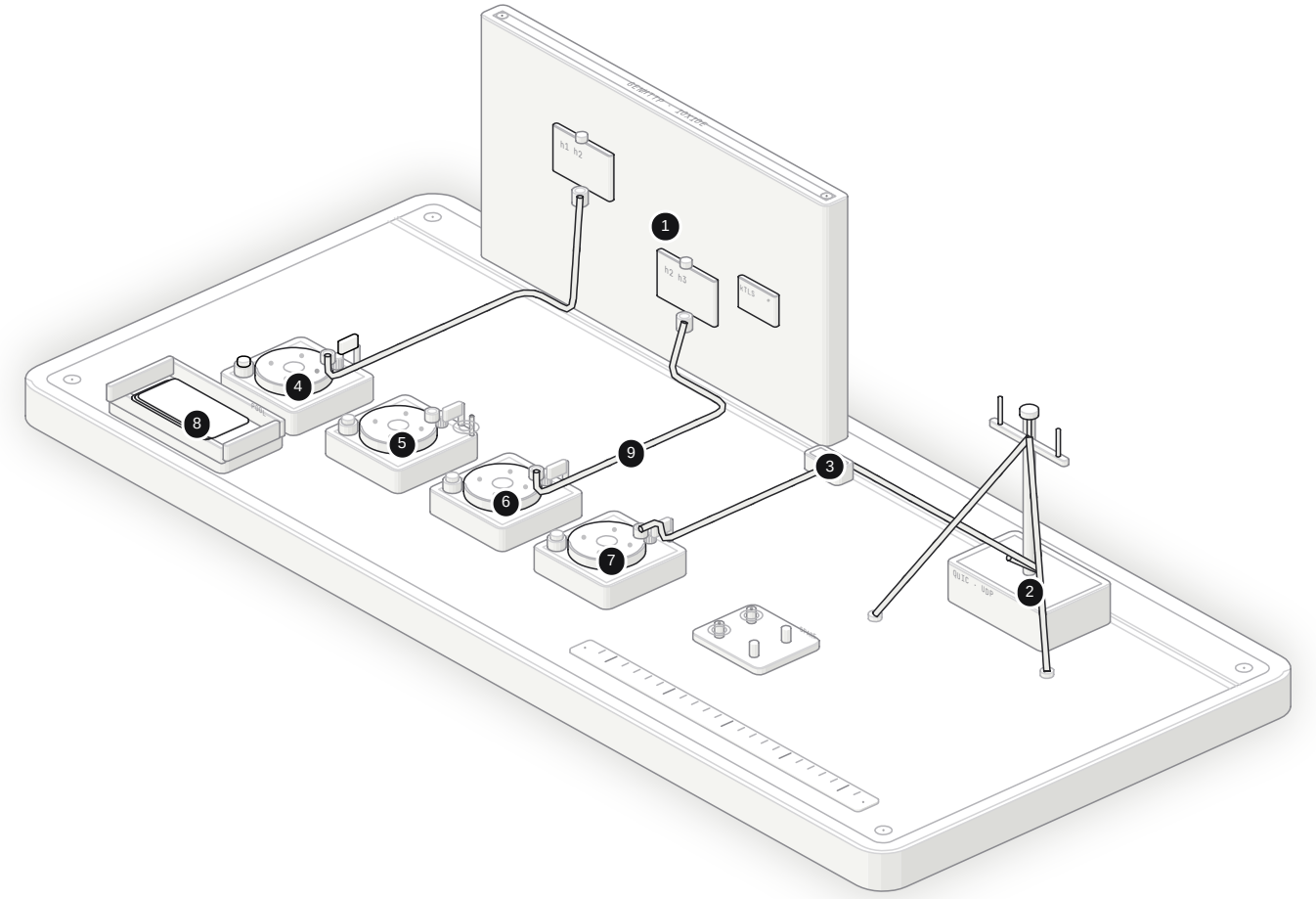
loxide

GHTTP-01
HOW IT WORKS
GENHTTP'S NATIVE LINUX ENGINE



How does a request wait for the network without a single syscall?

One reactor thread per core, each with its own io_uring ring of 8192 entries. Connections arrive on trunk ports — TCP with TLS and QUIC over UDP for HTTP/3 — and every reactor serves any of them. Each posted ticket is a submission; each lit one, a completion.

1 ring
IO_URING INSTANCE PER CORE

8192 entries
SUBMISSIONS AND COMPLETIONS PER RING

4096 slots
RECEIVE BUFFERS PER REACTOR, 32 KIB EACH

0 syscalls
SPENT WAITING FOR WORK TO ARRIVE

3 protocols
HTTP/1.1, HTTP/2 AND HTTP/3 ON ONE PORT

01 Trunk panel
Two listener ports, plain and TLS; the small block is kernel TLS.

02 Radio mast
QUIC over UDP — ngtcp2 and nghttp3 feed HTTP/3 in from the air.

03 Distribution block
Where cords meet: the junction the completions call back through.

04 Operator 1
Reactor on core 1, patching calls on its own ring.

05 Operator 2
Reactor on core 2; its cord rests on the hook between calls.

06 Operator 3
Reactor on core 3, fed from the second trunk port.

07 Operator 4
Reactor on core 4, fed through the distribution block.

08 Request pool
Request objects dealt from a tray, not allocated on the call path.

09 Patch cord
One cord per live connection, port to operator.